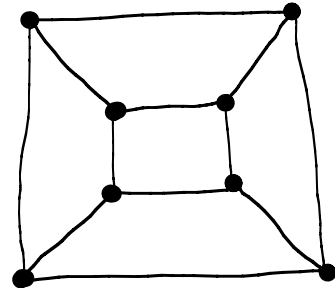
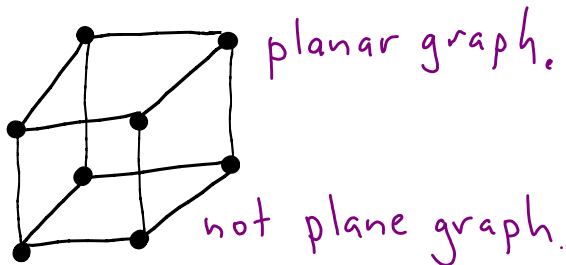


Chapter 4

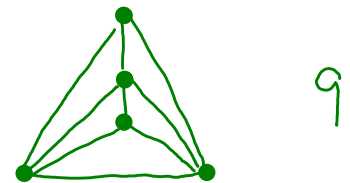
Planar Graphs

Definition 4.0.1. A graph is **planar** if it can be drawn in the plane with no edge crossings. Such a drawing is called a **plane graph**, and the graph is said to be **embedded** in the plane.

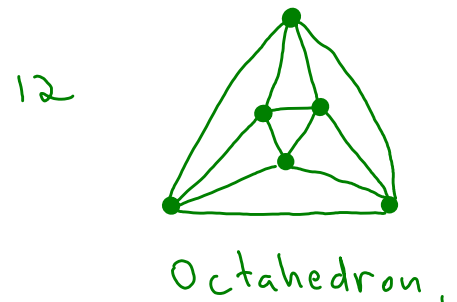


Exercises:

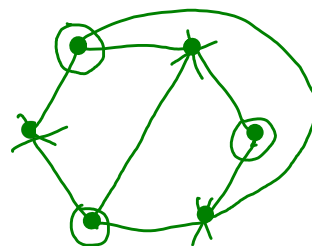
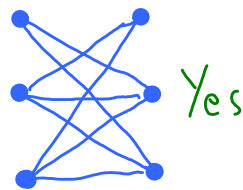
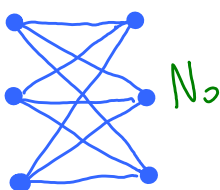
(a) What is the maximum number of edges in a simple 5-vertex planar graph?



(b) What is the maximum number of edges in a simple 6-vertex planar graph?

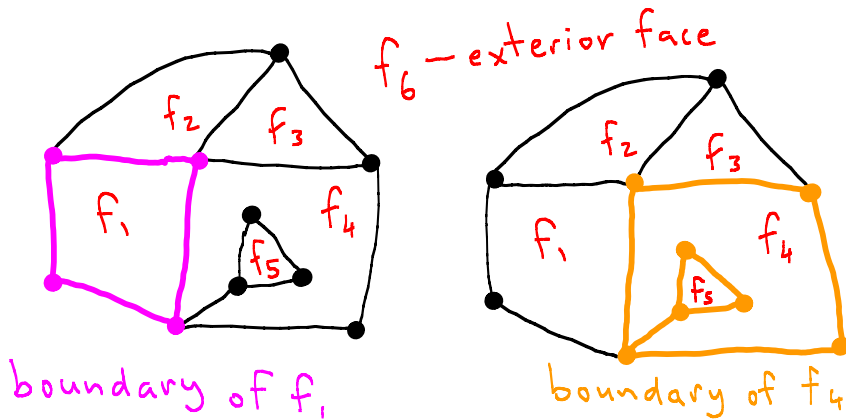


(c) Is $K_{3,3}$ planar? What about $K_{3,3}-e$?



$\exists$ 6-vertex planar graph with 12 edges, and 6-vertex non-planar graph with just 9 edges.

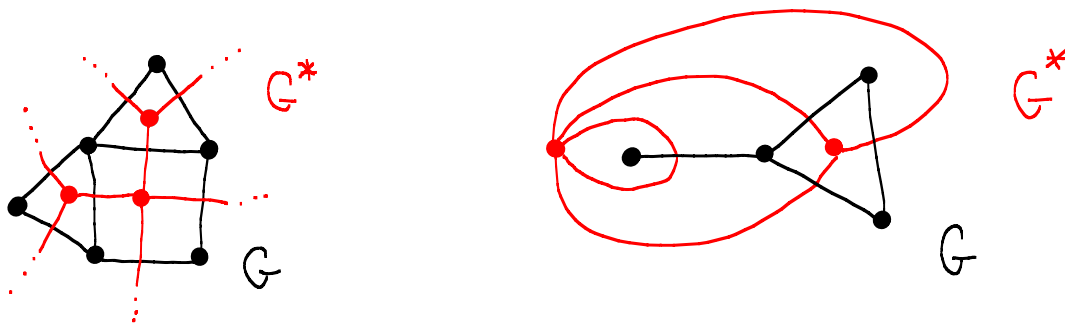
Definition 4.0.2. The **faces** of a plane graph are the resulting connected pieces of the plane that remain if the vertices and edges are removed. One of the faces is unbounded and is called the **exterior face**. The **boundary** of a face consists of the vertices and edges that are incident with the face. Two faces are **adjacent** or **neighbouring** if they are separated by a single edge, and the **length** of a face is the number of edges on its boundary plus the number of bridges on its boundary. The length of a face f is denoted by $\text{len}(f)$.



f_1 has length 4
 f_2, f_3, f_5 have length 3
 f_4 has length 9
 f_6 has length 6

A bridge in G is an edge such that $G - e$ has more components than G

Definition 4.0.3. The **dual graph** of a plane graph G is the plane graph obtained from G by placing a vertex in each face of G , and then for each edge e of G , joining the vertices in the two faces on either side of e by an edge that passes through e . If the same face is on both sides of e , then a loop results. The dual of a plane graph G is denoted by G^* .



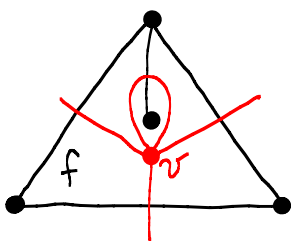
Theorem 4.0.4. For each vertex v of G^* , we have $\deg_{G^*}(v) = \text{len}(f)$ where f is the face of G corresponding to v . If F denotes the set of all faces of a plane graph G , then

$$\sum_{f \in F} \text{len}(f) = 2e(G).$$

Proof Let v be a vertex of G^* and let f be the corresponding face of G . For each edge e on the boundary of f , there is a corresponding edge e^* incident with v , and if e is a bridge, then there is a corresponding loop on v . Since loops count 2 towards the degree of a vertex, and bridges count two towards the length of a face, we have $\deg_{G^*}(v) = \text{len}(f)$.

Thus, if F denotes the set of faces of G , then we have

$$\sum_{f \in F} \text{len}(f) = \sum_{v \in V(G^*)} \deg(v) = 2e(G^*) = 2e(G).$$



$$\text{len}(f) = 5 \quad \deg(v) = 5$$

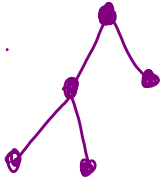
4.1 Euler's Identity and Kuratowski's Theorem

Theorem 4.1.1. (Euler's Identity) If a connected plane graph has n vertices, m edges, and r faces, then

$$n - m + r = 2.$$

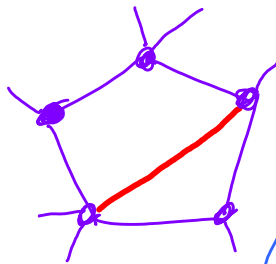
Proof: Induction on m . $m=0 \Rightarrow \bullet$ $\begin{matrix} n=1 \\ m=0 \\ r=1 \end{matrix} \checkmark$

Trees.



$$m=n-1, r=1 \checkmark$$

Otherwise.



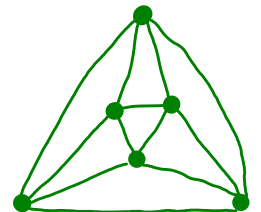
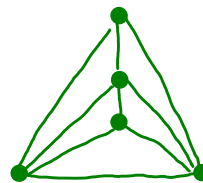
Deleting an edge

- decreases r by 1. $\checkmark$
 - decreases m by 1. $\checkmark$
 - leaves n unchanged.
- $\rightarrow n-m+r$ is unchanged. $\square$

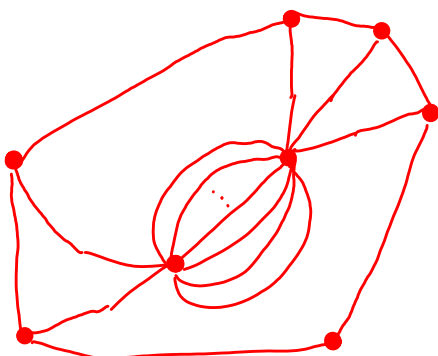
Theorem 4.1.2. If G is a simple n -vertex planar graph with $n \geq 3$, then $e(G) \leq 3n - 6$.

$$n=5: e(G) \leq 3n-6=9$$

$$n=6: e(G) \leq 3n-6=12$$



Exercise: Does Theorem 4.1.2 still hold if we omit "simple" or "with $n \geq 3$ "?



$$n=2: e(G) \leq 3n-6=0.$$

Theorem 4.1.2. If G is a simple n -vertex planar graph with $n \geq 3$, then $e(G) \leq 3n - 6$.

Proof It suffices to consider connected graphs because a new planar graph can be obtained by adding more edges to any disconnected graph. Since G is simple and has at least three vertices, every face has length at least 3. Let F be the set of faces of an embedding of G and let $r = |F|$. We have $2e(G) = \sum_{f \in F} \text{len}(f) \geq 3r$. Substituting $r \leq \frac{2}{3}e(G)$ into Euler's Identity, we obtain $e(G) \leq 3n - 6$.

$$n - m + r = 2 \Rightarrow n - m + \frac{2}{3}m \geq 2 \Rightarrow 3n - 3m + 2m \geq 6 \Rightarrow 3n - m \geq 6$$

$$\Downarrow$$

$$m \leq 3n - 6 \quad \square$$

Theorem 4.1.3. The graph K_5 is non-planar.

Proof: $n = 5 \Rightarrow 3n - 6 = 9$ but $e(K_5) = \binom{5}{2} = 10$.

Theorem 4.1.4. The graph $K_{3,3}$ is non-planar.

$$3n - 6 = 12 \quad \text{and} \quad e(K_{3,3}) = 9 \quad \text{but} \dots$$

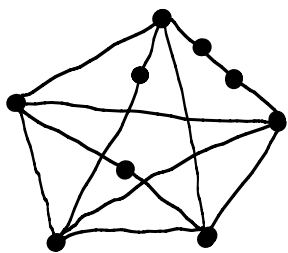
Suppose Planar.

$$\text{Euler's Identity, } n - m + r = 2 \Rightarrow 6 - 9 + r = 2 \Rightarrow r = 5.$$

$K_{3,3}$ has no 3-cycles so each face has at least 4 edges.

$$\text{So } 18 = 2m = \sum_{f \in F} \text{len}(f) \geq 5 \times 4 = 20. \quad \text{Contradiction.}$$

Definition 4.1.5. Any graph obtained from a graph G by replacing edges with pairwise internally disjoint paths is called a **subdivision** of G .



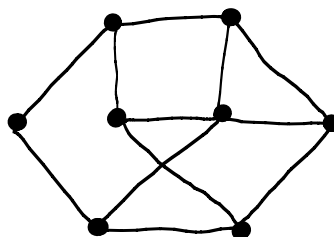
A subdivision
of K_5

A subdivision of G is
planar if and only if
 G is planar.

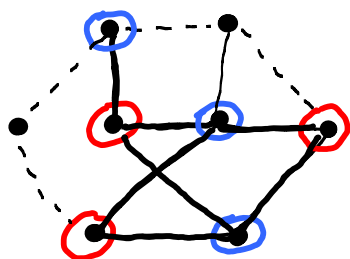
Any graph containing a subdivision of K_5 or $K_{3,3}$ is non-planar.

Theorem 4.1.6. (Kuratowski's Theorem) A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$.

Exercise: Is this graph planar?



No. It contains a subdivision of $K_{3,3}$.



subdivided edges of $K_{3,3}$ subgraph
are dotted.

A graph containing a subdivision of K_5 has at least 5 vertices
of degree at least 4.

4.2 The Four Colour Theorem

Note: In this section "graph" means "simple graph".

The Four Colour Problem asks ...

Is it true that the countries of any map can be coloured with four colours so that any two neighbouring countries are assigned distinct colours?

The Four Colour Theorem states that four colours are indeed sufficient.

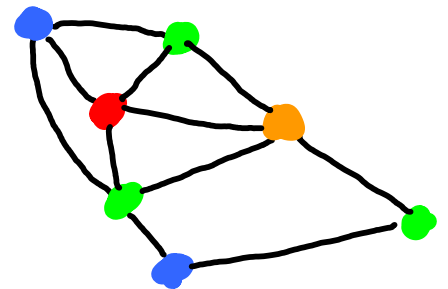
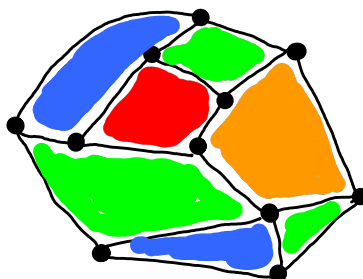
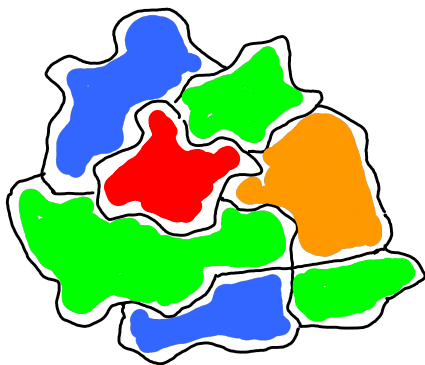
See historical overview of 4CT in the notes:—

- originated ~ 1850 .
- solved in 1976 by Appel and Haken (University of Illinois).
- notorious for "false proofs".
- long, tedious, complicated proof — using computers.

4CT $\longleftrightarrow$ The faces of any plane graph can be coloured with 4 colours such that adjacent faces are assigned distinct colours.

$\updownarrow$ dual graph

Every planar graph is 4-colourable

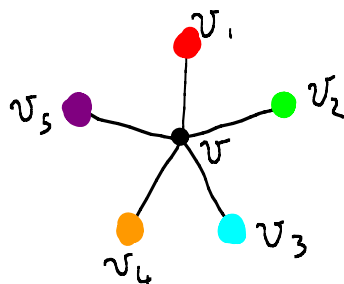


A **colouring** of a graph is an assignment of colours to its vertices, and a colouring with k or fewer colours is a **k -colouring**. A k -colouring is **proper** if adjacent vertices are assigned distinct colours, and a graph is k -colourable if it has a proper k -colouring. The term **(vertex) colour class c** refers to the set of vertices of colour c .

Theorem 4.2.1. (Five Colour Theorem) Every planar graph is 5-colourable.

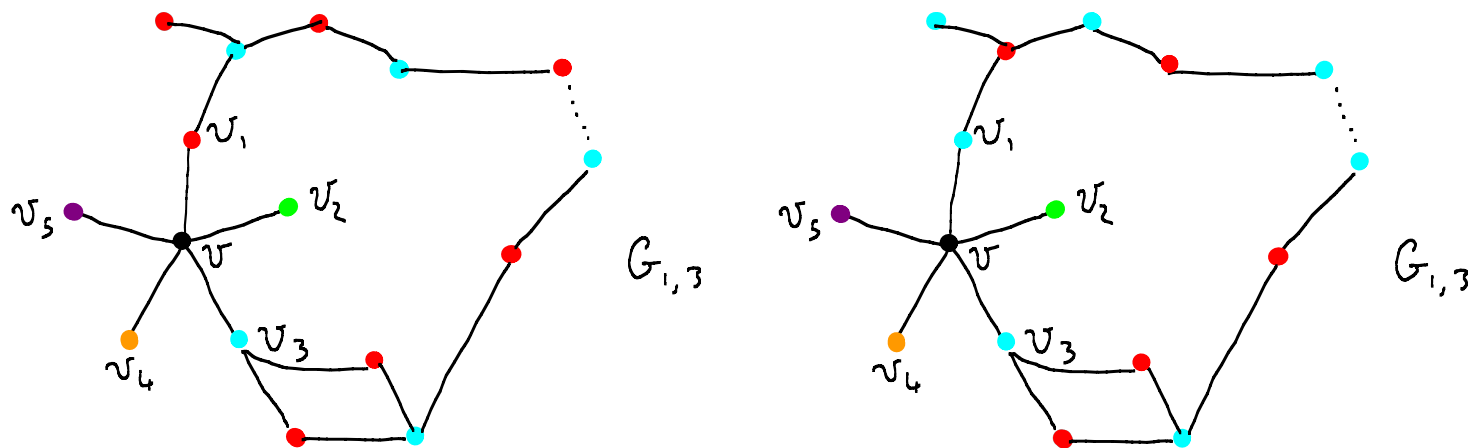
Proof Suppose the theorem is false and let G be a counter-example of minimum order. Let n be the order of G . Every graph with 5 or fewer vertices is 5-colourable, so $n \geq 6$. If every vertex of G has degree at least 6, then G has at least $3n$ edges. But this contradicts Theorem 4.1.2 and we conclude that G has a vertex v of degree at most 5.

The graph $G - v$ is 5-colourable by the minimality of G , so we can give $G - v$ a proper 5-colouring. Each of the 5 colours must occur on a neighbour of v because otherwise there is a colour available for vertex v . Thus, v has degree 5 and the 5 neighbours of v are assigned distinct colours. Let

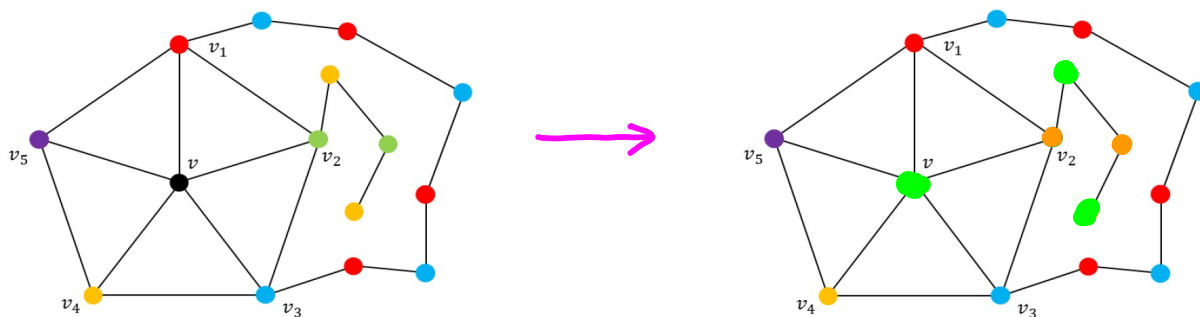


v_1, v_2, v_3, v_4, v_5 be the five neighbours of v in clockwise order around v , and without loss of generality for $i = 1, 2, 3, 4, 5$, let i be the colour of v_i .

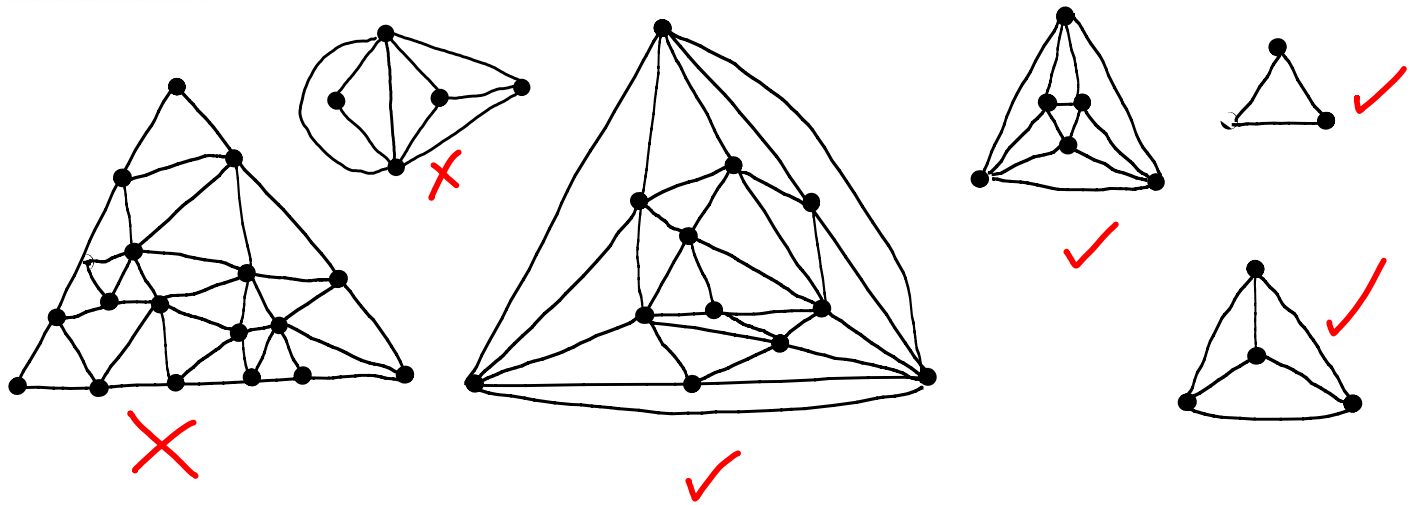
Let $G_{1,3}$ be the subgraph of $G - v$ induced by the vertices of colour 1 and 3. If v_1 and v_3 are in different components of $G_{1,3}$, then we can switch the colours on the component of $G_{1,3}$ that contains v_1 and then produce a proper 5-colouring of G by assigning colour 1 to v . Thus, v_1 and v_3 are in the same component of $G_{1,3}$, and there exists a v_1, v_3 -path all of whose vertices are coloured 1 or 3. This path can be completed to a cycle C by adding the vertex v and the edges vv_1 and vv_3 .



One of the vertices v_2 and v_4 is in the interior of C and the other is in the exterior. Thus, v_2 and v_4 are in different components of $G_{2,4}$, where $G_{2,4}$ is the subgraph of $G - v$ induced by the vertices of colour 2 and 4. Thus, if we switch the colours on the component of $G_{2,4}$ that contains v_2 and assign colour 2 to v , then we obtain a proper 5-colouring of G . This is a contradiction and we conclude that every planar graph is 5-colourable. $\square$



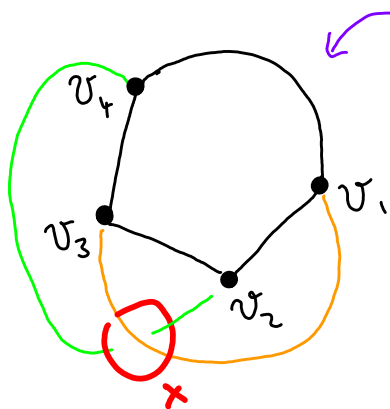
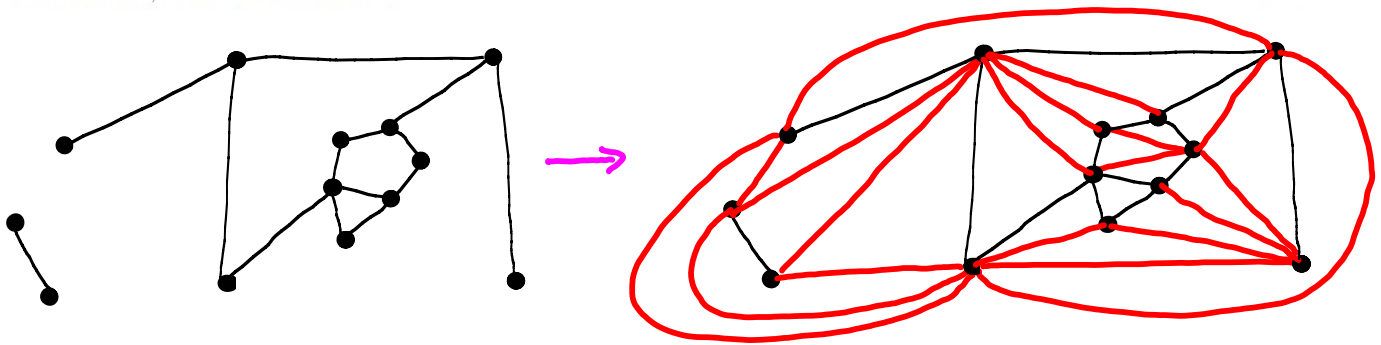
Definition 4.2.2. A planar graph in which the boundary of every face is a triangle is called a **planar triangulation**.



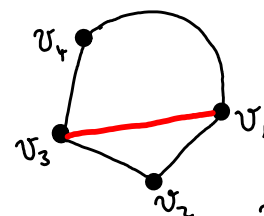
Theorem 4.2.3. Every planar graph is 4-colourable if and only if every planar triangulation is 4-colourable.

4CT $\leftrightarrow$ Every planar triangulation is 4-colourable.

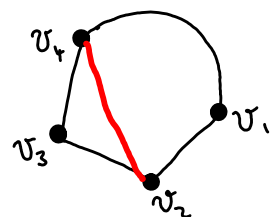
Proof If every planar graph is 4-colourable, then certainly every planar triangulation is 4-colourable. To prove the converse, suppose that every planar triangulation is 4-colourable and let G be a plane graph. If G is not a triangulation, then we can add edges to G until we obtain a planar triangulation. By assumption, this planar triangulation has a proper 4-colouring, and this is also a proper 4-colouring of G . Thus, G is 4-colourable. $\square$



m -cycle face with $m \geq 4$
 v_1v_3 and v_2v_4 are not both edges. $\Rightarrow$

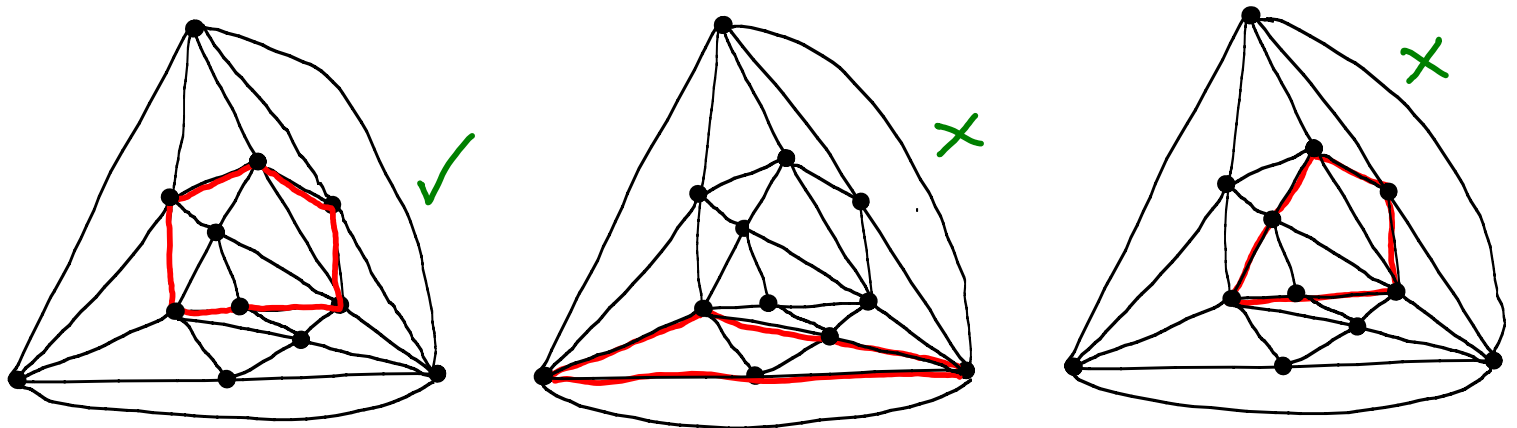


OR



Definition 4.2.4. In the context of the Four Colour Problem, a **minimal counter-example** is a planar triangulation G such that G is not 4-colourable, and such that every planar triangulation with fewer vertices than G is 4-colourable.

Definition 4.2.5. A **separating cycle** in a planar triangulation is a cycle C such that if the vertices of C together with any edges incident with any of them are deleted, then the resulting graph has at least two connected components.

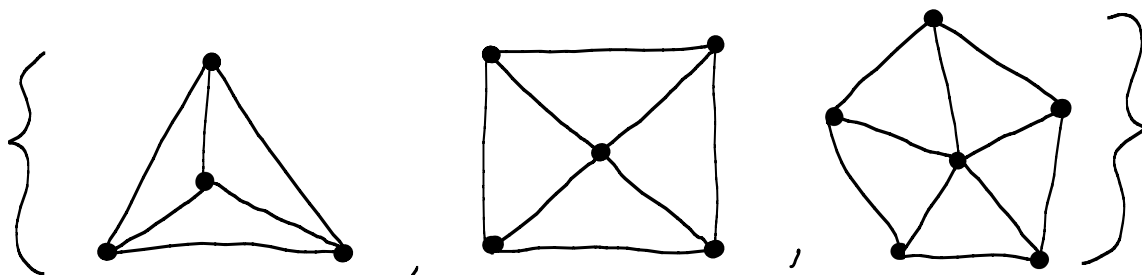


Which of these is a separating cycle?

Definition 4.2.6. A **configuration** in a planar triangulation consists of a separating cycle C , called the **ring**, together with the part of the graph inside C . A set S of configurations is **unavoidable** if any minimal counter-example must contain a copy of at least one configuration from S . A configuration is **reducible** if a planar triangulation containing it cannot be a minimal counter-example.

If there exists an unavoidable set of reducible configurations, then every planar graph is 4-colourable.

In the context of the Five-Colour Theorem, we have the following unavoidable set of reducible configurations.



How did we show:
- unavoidable?
- reducible?

Discharging: a method for proving sets of configurations are unavoidable.

Theorem 4.2.7. If G is a planar triangulation, then

$$\sum_{v \in V(G)} (6 - \deg(v)) = 12.$$

Proof Let F be the set of faces of a planar triangulation and let $r = |F|$. Then

$$\sum_{v \in V(G)} \deg(v) = 2e(G) = \sum_{f \in F} \text{len}(f) = 3r.$$

But by Euler's Identity we have $6n(G) - 6e(G) + 6r = 12$, so

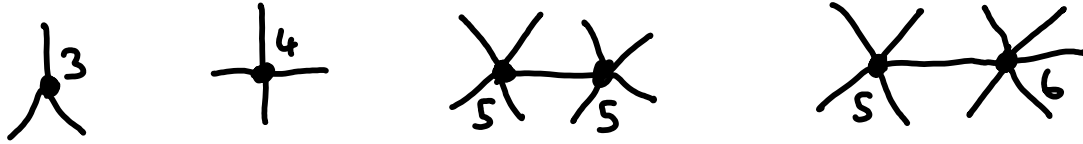
$$12 = 6n(G) - 3 \sum_{v \in V(G)} \deg(v) + 2 \sum_{v \in V(G)} \deg(v) = 6n(G) - \sum_{v \in V(G)} \deg(v) = \sum_{v \in V(G)} (6 - \deg(v)).$$

Exercise: Use Theorem 4.2.7 to show that a planar triangulation with minimum degree 5 and maximum degree 7 has at least 13 vertices of degree 5.

$$\sum_{v \in V(G)} (6 - \deg(v)) = 12$$

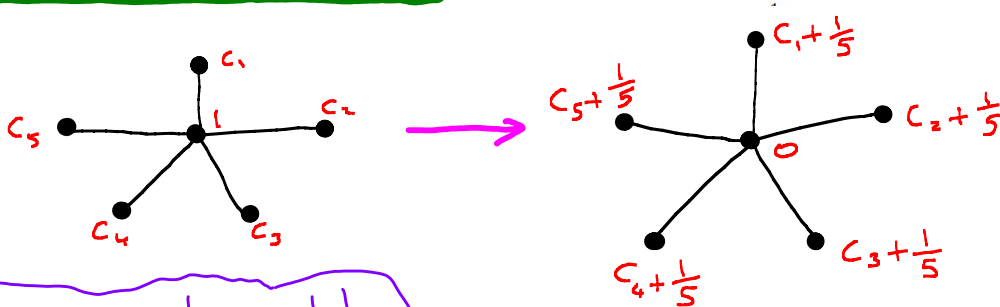
There exists a vertex of degree 7 which contributes -1 to the sum. Vertices of degree 6 contribute 0. Thus, in order for the sum to be 12, there exist at least 13 vertices of degree 5 (because there are no vertices of degree less than 5).

Theorem: Every planar triangulation contains a vertex of degree 3, a vertex of degree 4, neighbouring vertices both of degree 5, OR neighbouring vertices of degree 5 and 6.



Proof For a contradiction, suppose there is a planar triangulation G containing none of these. Thus, there are no vertices of degree 3 or 4, and the neighbours of any vertex of degree 5 have degree at least 7. Define the *charge* on each vertex v to be $6 - \deg(v)$. Then Theorem 4.2.7 says that the sum of the vertex charges, the *net charge*, in G is 12.

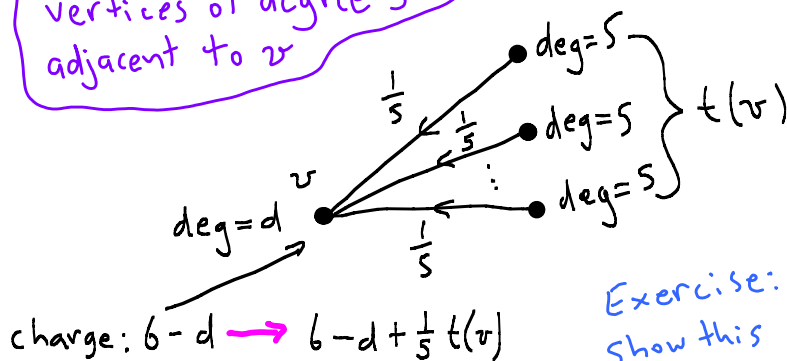
Redistribute the charge by shifting $\frac{1}{5}$ of the charge on each vertex v of degree 5 to each of the 5 neighbours of v . The net charge is unchanged and each vertex of degree 5 now has charge zero. Each vertex of degree 6 still has charge zero.



Where do we get positive charge ???

(in order to get 12)

For each vertex v , let $t(v)$ be the number of vertices of degree 5 adjacent to v



If $\deg(v) = 8$, then

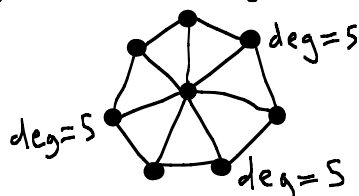
charge on v is

$$-2 + \frac{1}{5} t(v) \leq -2 + \frac{8}{5} < 0.$$

Exercise: show this

$\Rightarrow$ Similarly, if $\deg(v) \geq 9$, then charge on v is < 0 .

If $\deg(v) = 7$, then charge on $v \leq 0$ unless $t(v) \geq 6$ ($6 - 7 + \frac{5}{5} = 0$).



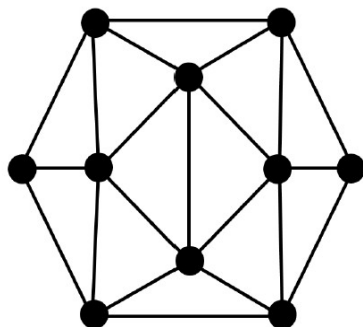
Net charge ≤ 0 .

Contradiction.

□

Throughout the early to mid 1900s, methods of increasing complexity were used to construct reducible configurations and unavoidable sets.

In 1913, George Birkhoff, who is famous in many areas of mathematics and particularly well known for his work on ergodic theory and the three-body problem, proved that any configuration with at least seven vertices whose ring is a 5-cycle is reducible. He also proved that what is now known as the *Birkhoff diamond* (shown below) is reducible.



Proofs of reducibility of configurations are long and complicated. One of the difficulties is the rapid increase in the number of distinct possible colourings of the ring as the size (number of edges) of the ring increases. Each of these colourings needs to be considered in reducibility proofs. The number of distinct colourings for rings of sizes 6 to 14 is given in the table below [25].

ring size	6	7	8	9	10	11	12	13	14
number of colourings	31	91	274	820	2461	7381	22144	64430	199291

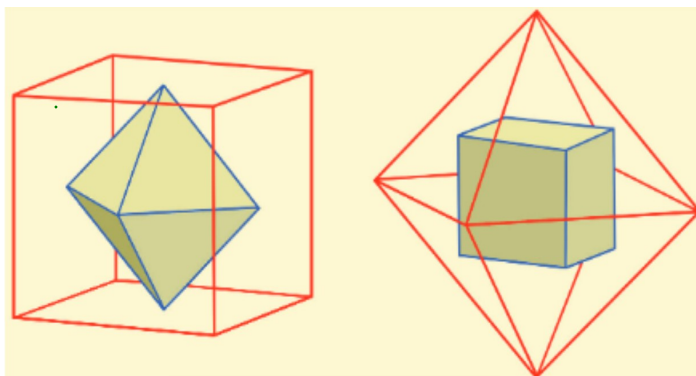
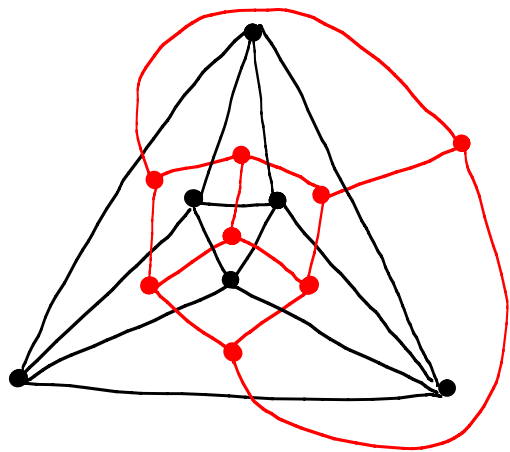
By the end of the 1960s, Haken and others had been using computers extensively to help with the huge number of cases that needed to be considered in the *unavoidable set of reducible configurations* approach to the Four Colour Problem. In late June, they constructed an unavoidable set of 1936 configurations with ring sizes at most 14. A couple of days later they had proved that the set was also reducible.

In the 1977 published version of the Appel and Haken's proof [1], the number of reducible configurations in their unavoidable set had been reduced from 1936 to 1482. The later proof of Robertson, Sanders, Seymour and Thomas [14], which was published in 1997, used an unavoidable set of 633 configurations.

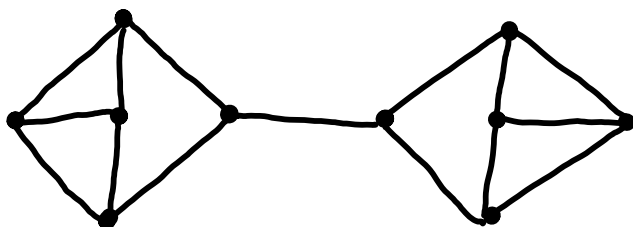
4.3 Snarks

We have seen (see Theorem 4.2.3) that the Four Colour Theorem is equivalent to the statement that every planar triangulation is 4-colourable. The dual of a planar triangulation is a simple 2-edge-connected 3-regular planar graph. Thus, the Four Colour Theorem is also equivalent to the statement that the faces of every simple 2-edge-connected 3-regular planar graph can be coloured with 4 colours so that adjacent faces are assigned distinct colours.

no bridges



Exercise: Draw a simple connected planar 3-regular graph that is not 2-edge-connected.



We have similar definitions for face colourings of plane graphs as we have for vertex and edge colourings of graphs.

Definition 4.3.1. A **face colouring** of a plane graph is an assignment of colours to its faces, and a face colouring with k or fewer colours is a **k -face colouring**. A k -face colouring is **proper** if adjacent faces are assigned distinct colours, and a plane graph is **k -face-colourable** if it has a proper k -face colouring.

Compare with similar definitions for vertex and edge colourings:-

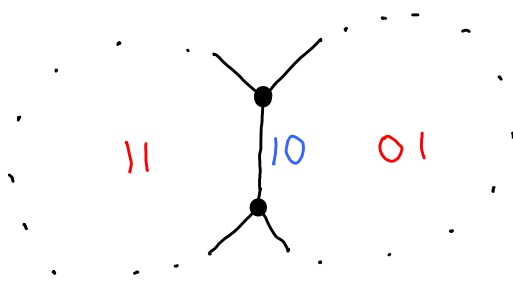
A **colouring** of a graph is an assignment of colours to its vertices, and a colouring with k or fewer colours is a **k -colouring**. A k -colouring is **proper** if adjacent vertices are assigned distinct colours, and a graph is k -colourable if it has a proper k -colouring. The term **(vertex) colour class c** refers to the set of vertices of colour c .

An **edge colouring** of a graph is an assignment of colours to its edges, and an edge colouring with k or fewer colours is a **k -edge colouring**. A k -edge colouring is **proper** if adjacent edges are assigned distinct colours, and a graph is **k -edge-colourable** if it has a proper k -edge colouring. The term **(edge) colour class c** refers to the set of edges of colour c . For edge colourings, we sometimes we use the term **colour class c** to mean the subgraph induced by colour class c .

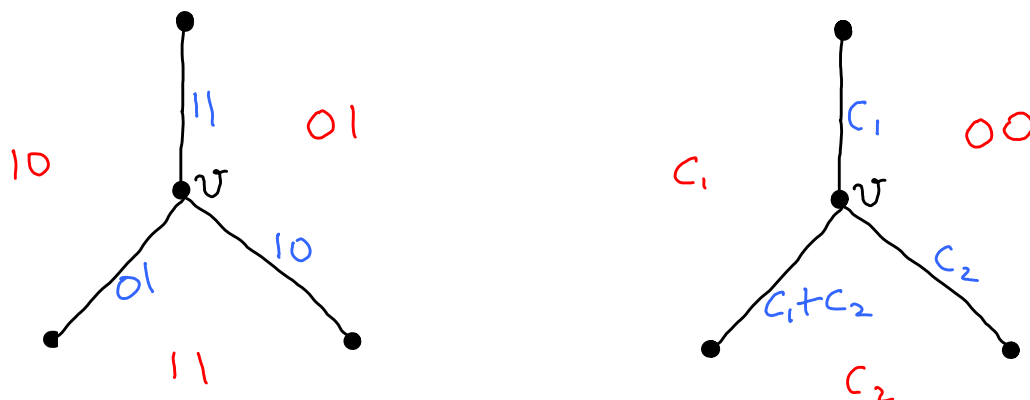
The following result, which was proved by Tait in 1878 [16], shows that the Four Colour Theorem is also equivalent to the statement that every simple 2-edge-connected 3-regular plane graph is 3-edge-colourable.

Theorem 4.3.2. Let G be a simple 2-edge-connected 3-regular planar graph. Then G is 4-face-colourable if and only if G is 3-edge-colourable.

Proof First suppose that G has a proper 4-face colouring. Let the four colours be 00, 01, 10 and 11, the elements of the elementary abelian 2-group $\mathbb{Z}_2 \times \mathbb{Z}_2$. Colour each edge e of G with the sum of the colours of the faces that e separates. We claim that this produces a proper 3-edge colouring of G with colours 01, 10 and 11.

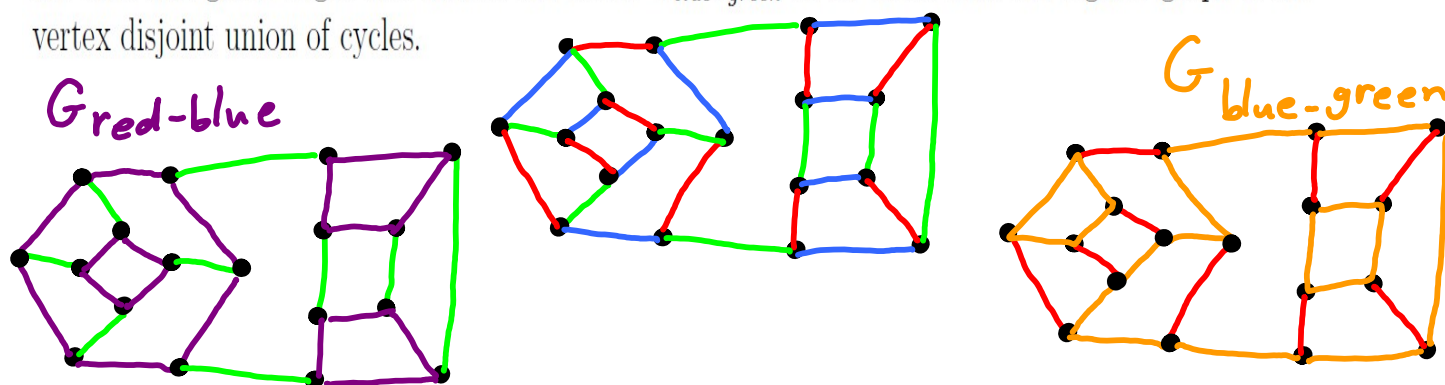


To see this, first observe that 00 never occurs as each edge separates faces of distinct colours. If the colours on the pairwise adjacent faces around a vertex v are 01, 10 and 11, then it follows that these same 3 colours occur on the three edges incident with v (because $01 + 10 = 11$, $01 + 11 = 10$ and $10 + 11 = 01$).

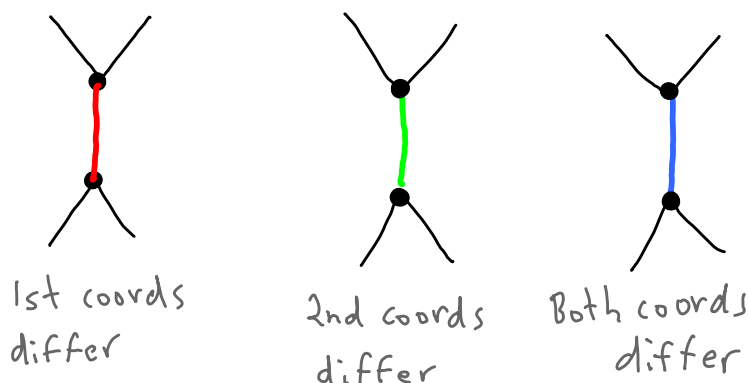
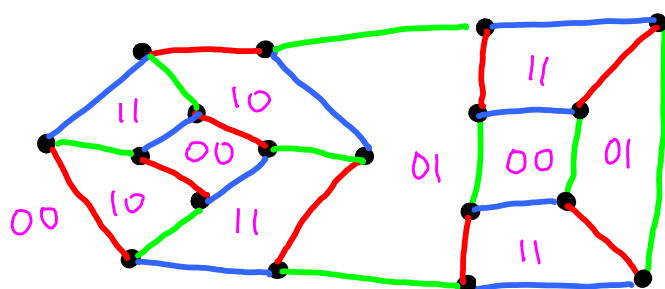


If the colours on the pairwise adjacent faces around a vertex v are 00, c_1 and c_2 , where $c_1, c_2 \in \{01, 10, 11\}$, then it follows that the two edges incident with v and on the boundary of the face coloured 00 are assigned colours c_1 and c_2 and the third edge incident with v is assigned colour $c_1 + c_2$, which is the third element of $\{01, 10, 11\}$. Thus, we have a proper 3-edge colouring of G .

For the converse, now suppose that G has a proper 3-edge colouring. Let these colours be red, blue and green. Since G is 3-regular, the red and blue edges induce a 2-factor $G_{\text{red-blue}}$ of G . Similarly, the blue and green edges also induce a 2-factor $G_{\text{blue-green}}$ of G . Note that a 2-regular graph is the vertex disjoint union of cycles.

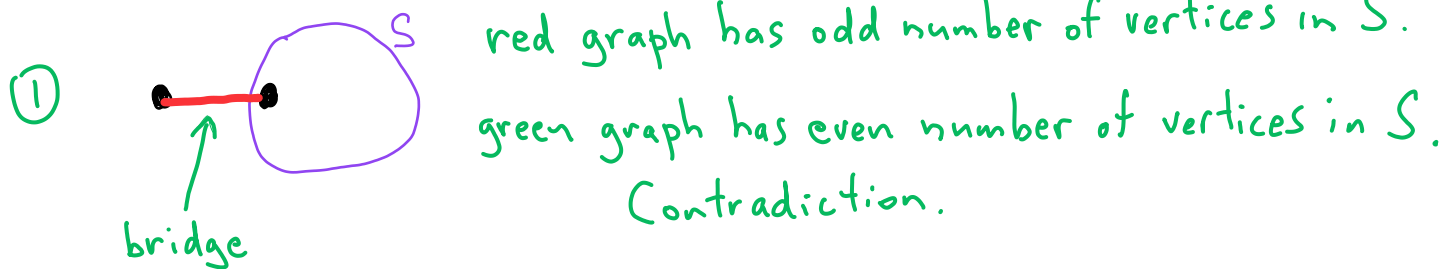


We assign colours from $\{00, 01, 10, 11\}$ to the faces of G as follows. The first coordinate of the colour of a face is the parity (0 for even, 1 for odd) of the number of cycles of $G_{\text{red-blue}}$ containing the face, and the second coordinate is the parity of the number of cycles of $G_{\text{blue-green}}$ containing the face. We claim that this is a proper face colouring of G .



Exercise: Show that any connected 3-edge-colourable 3-regular graph is 2-edge-connected.

Suppose a 3-regular graph with a bridge has a proper 3-edge colouring ...



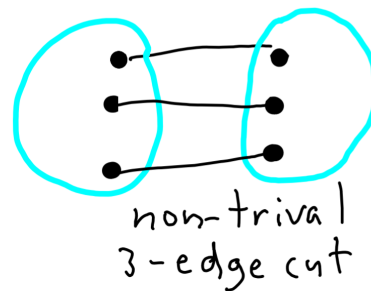
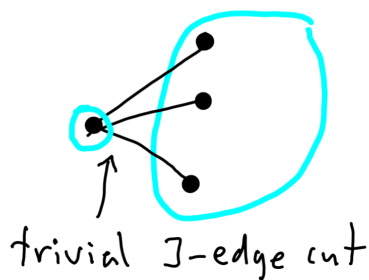
② The union of any two colour classes is a 2-factor, but a bridge is in no cycles. Contradiction.

Since the Four Colour Theorem is equivalent to the statement that every simple 2-edge-connected 3-regular planar graph is 3-edge-colourable, simple 2-edge-connected 3-regular graphs that are not 3-edge-colourable (these are necessarily non-planar by the Four Colour Theorem) are of interest. Such graphs are called *snarks*.

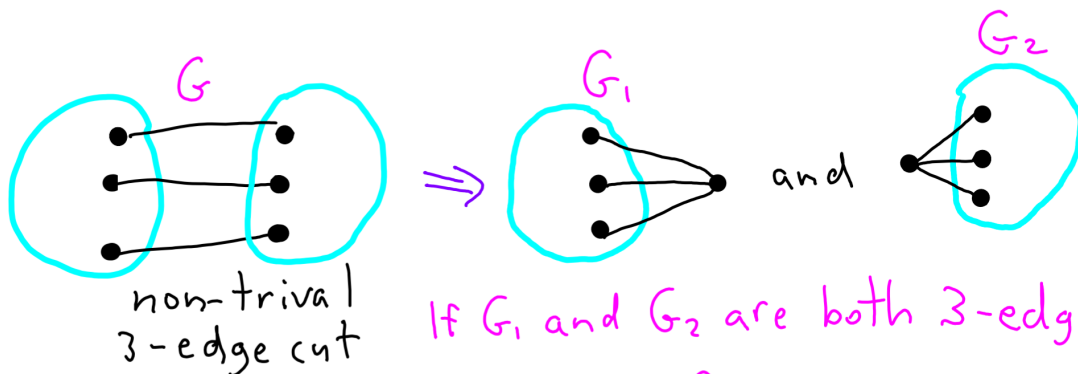
Definition 4.3.3. A **snark** is a simple 2-edge-connected 3-regular graph that is not 3-edge-colourable.

Recall that the Four Colour Theorem is equivalent to the statement that every simple 2-edge-connected 3-regular planar graph is 3-edge-colourable.

We will now show that the Four Colour Theorem is equivalent to the statement that every simple 3-connected 3-regular planar graph with girth 5, and with no non-trivial 3-edge-cuts, is 3-edge-colourable.

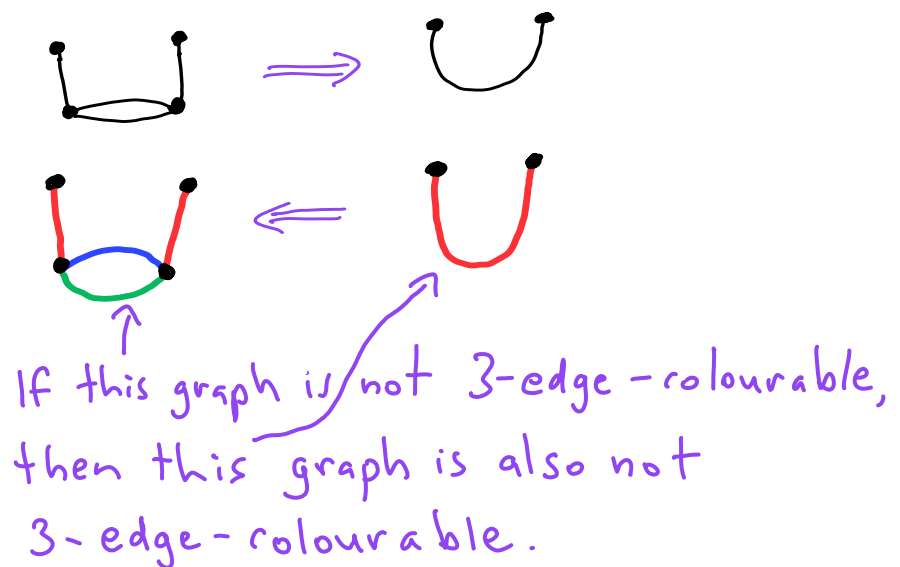
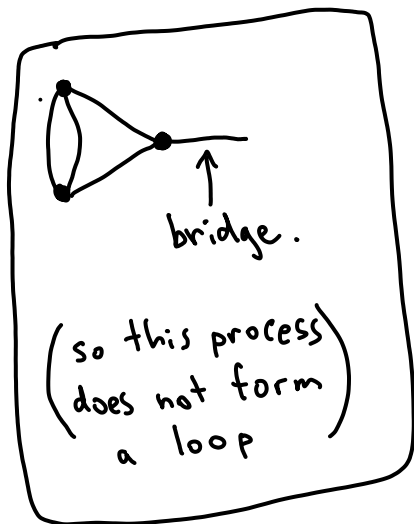


Exercise: Show how to construct a smaller non-3-edge-colourable 3-regular graph from a non-3-edge-colourable 3-regular graph with a non-trivial 3-edge-cut.



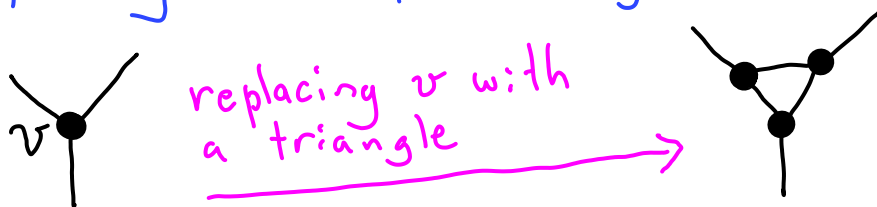
If G_1 and G_2 are both 3-edge-colourable, then so is G .

If there is a 2-edge-connected 3-regular graph with a double edge that is not 3-edge-colourable, then we can replace the double edge, and the two other edges that are adjacent to the double edge, with a single edge to form a smaller 2-edge-connected 3-regular graph that is not 3-edge-colourable.



Exercises:

- ① Let v be a vertex in a simple 3-regular graph G , and let G' be the graph obtained from G by "replacing v with a triangle".

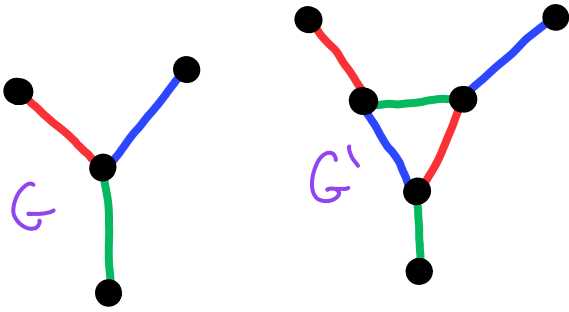


Convince yourself that G is a snark if and only if G' is a snark.

- ② Draw a snark with 10 vertices.
- ③ Draw a snark with 14 vertices.
- ④ Show that if there exists a non-3-edge-colourable 3-regular graph with n vertices and girth 4, then there exists a non-3-edge colourable 3-regular graph with $n-4$ vertices

Solutions:

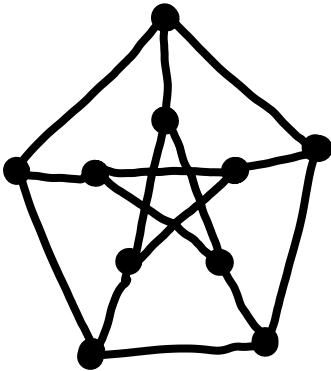
①



G is 3-edge-colourable
iff G' is 3-edge-colourable.

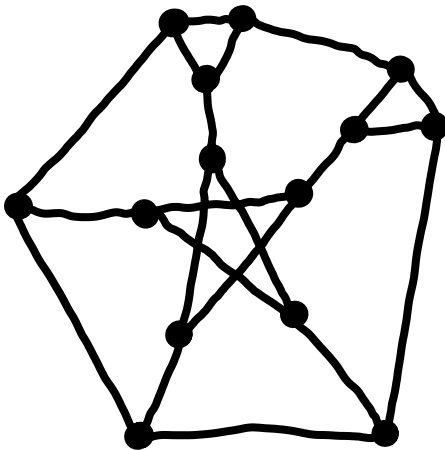
G is 2-edge-connected
iff G' is 2-edge-connected.

②



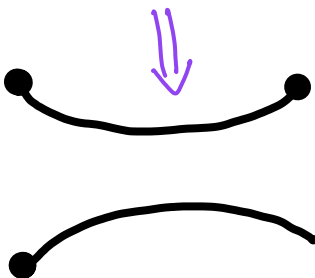
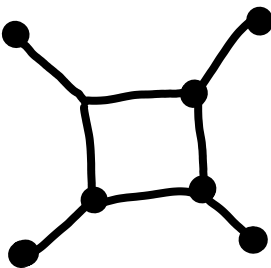
The Petersen graph is a snark
with 10 vertices.

③

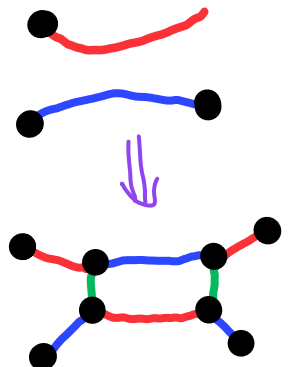
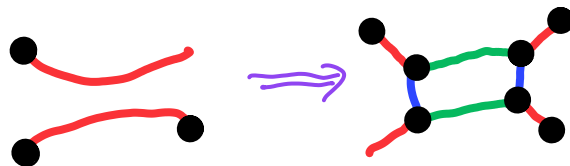


A snark with 14 vertices
can be constructed from the
Petersen graph by replacing
two vertices with triangles.

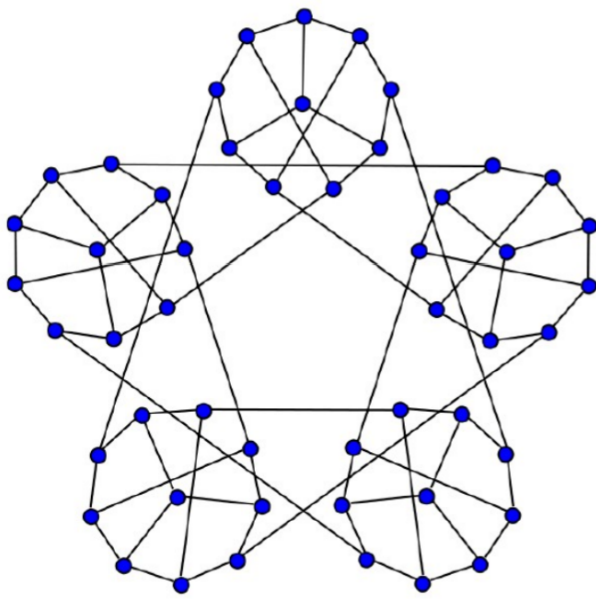
④



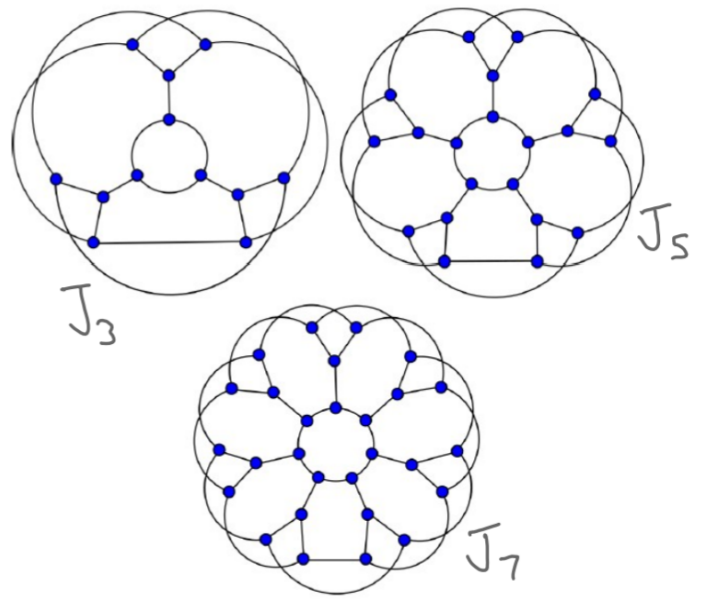
new 3-regular
graph with
 $n-4$ vertices
(girth 4 $\Rightarrow$ no loops)



The Petersen graph is the smallest snark. Shown below is the Szekeres snark (at left) and the Flower snarks J_3 , J_5 and J_7 (at right).



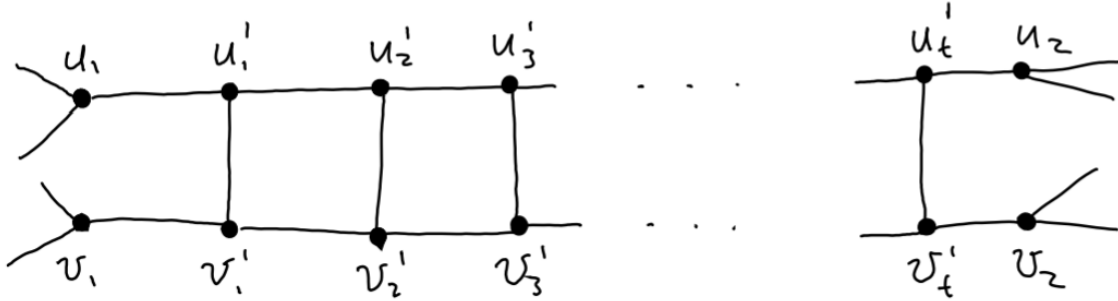
Szekeres snark



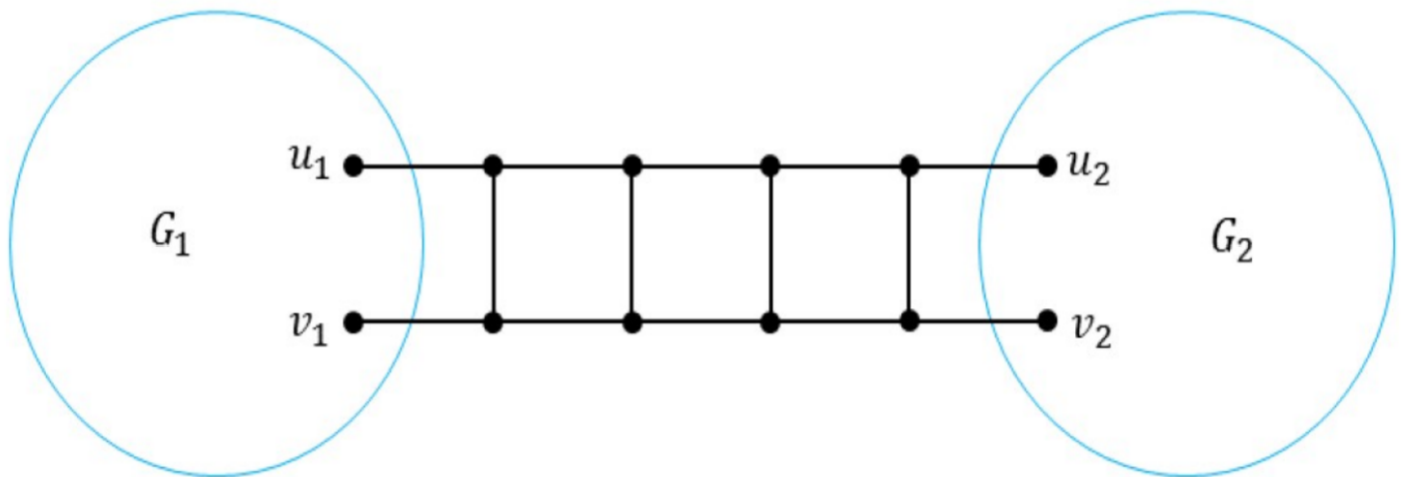
Still want to show that we can "replace"
 2-edge-connected with 3-connected.
 We do this in the next section.

4.4 Grinberg's Theorem and the Tutte Graph

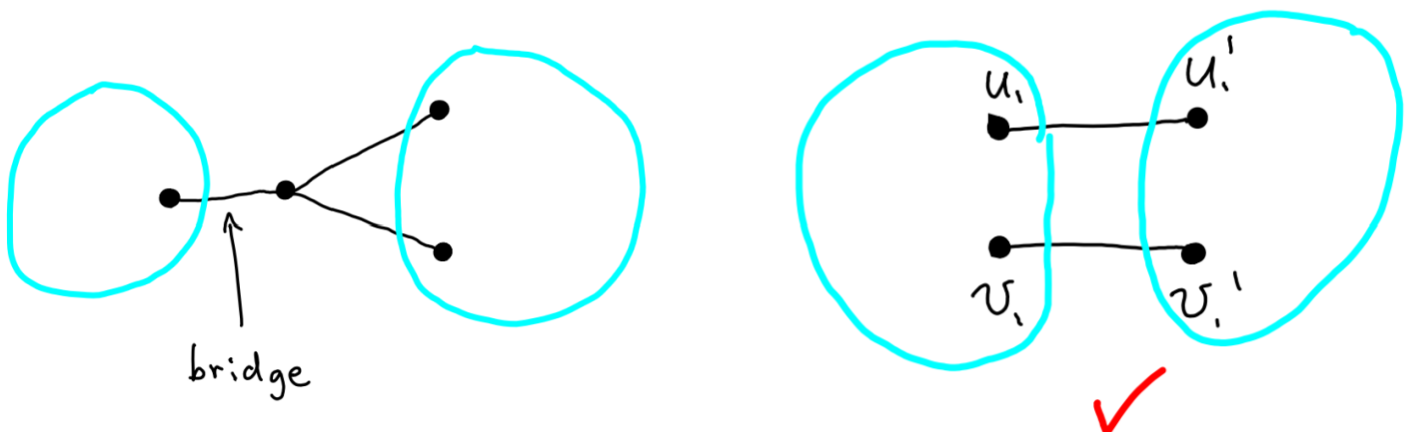
Definition 4.4.1. Let G be a connected 3-regular graph and let u_1, u_2, v_1, v_2 be distinct vertices of G . A **ladder of length $t + 1$ in G joining (u_1, v_1) to (u_2, v_2)** consists of disjoint paths $[u_1, u'_1, u'_2, \dots, u'_t, u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v_2]$ and edges $u'_1 v'_1, u'_2 v'_2, \dots, u'_t v'_t$ such that $\{u_1 u'_1, v_1 v'_1\}$ is an edge cut of G .



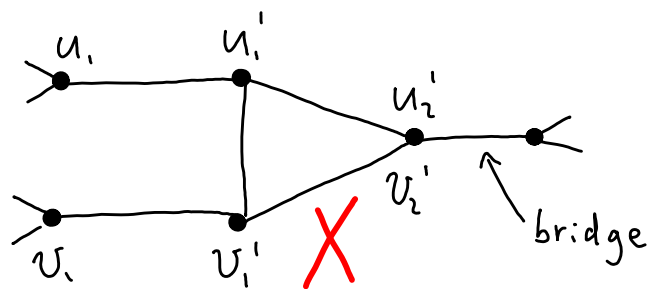
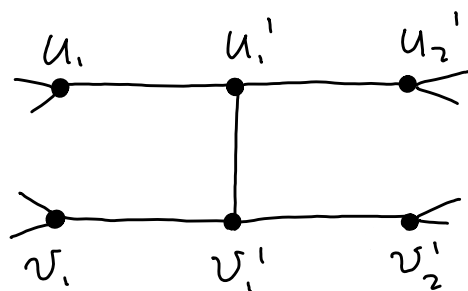
Lemma 4.4.2. Let G be a 3-regular graph with edge connectivity 2. Then G consists of vertex-disjoint subgraphs G_1 and G_2 , non-adjacent vertices $u_1, v_1 \in V(G_1)$, non-adjacent vertices $u_2, v_2 \in V(G_2)$, and a ladder joining (u_1, v_1) to (u_2, v_2) .



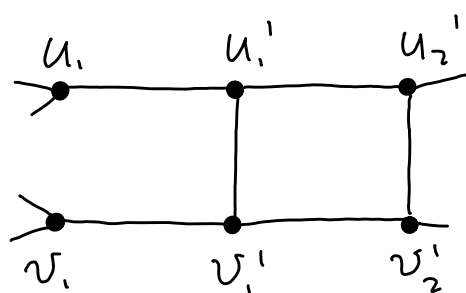
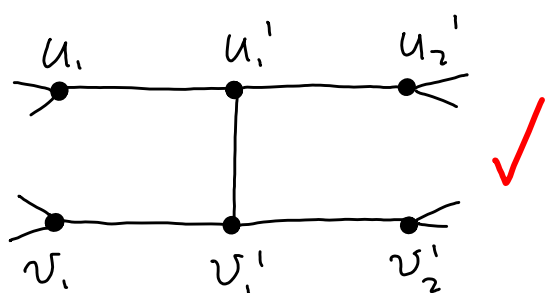
Proof If G has a 2-edge cut consisting of adjacent edges, then the third edge on the common vertex where they are incident is a bridge, contradicting the fact that G has edge connectivity 2. Thus, there exists a 2-edge cut $\{u_1 u'_1, v_1 v'_1\}$ where u_1, v_1 are distinct and u'_1, v'_1 are distinct. If u_1, v_1 and u'_1, v'_1 are non-adjacent, then we have the desired ladder of length 1 consisting of the paths $[u_1, u'_1 = u_2]$ and $[v_1, v'_1 = v_2]$.



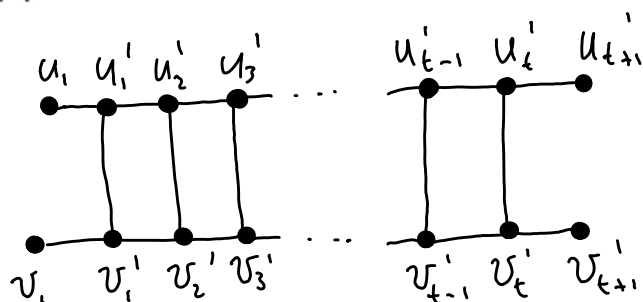
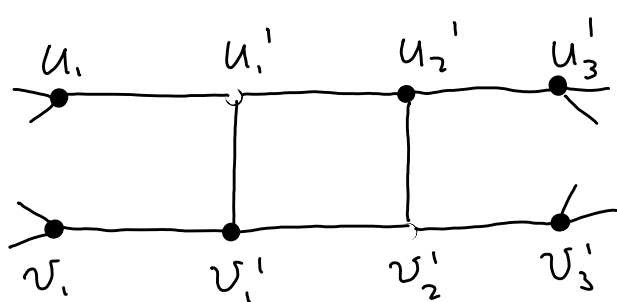
If u_1, v_1 are non-adjacent and u'_1, v'_1 are adjacent, then we let u'_2 be the third neighbour of u'_1 (the other two neighbours being u_1 and v'_1) and let v'_2 be the third neighbour of v'_1 (the other two



neighbours being v_1 and u'_1). If $u'_2 = v'_2$, then the third edge incident on this vertex is a bridge. So u'_2 and v'_2 are distinct. If u_1, v_1 and u'_2, v'_2 are non-adjacent, then we have the desired ladder of length 2 consisting of the paths $[u_1, u'_1, u'_2 = u_2]$ and $[v_1, v'_1, v'_2 = v_2]$ and the edge $u'_1 v'_1$.



If u_1, v_1 are non-adjacent and u'_2, v'_2 are adjacent, then we let u'_3 be the third neighbour of u'_2 (the other two neighbours being u'_1 and v'_2) and let v'_3 be the third neighbour of v'_2 (the other two neighbours being v'_1 and u'_2), and continue “extending the ladder” until we reach non-adjacent vertices u'_{t+1} and v'_{t+1} , so that we have the required ladder consisting of disjoint paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ and the edges $u'_1 v'_1, u'_2 v'_2, \dots, u'_t v'_t$.



We can repeat the same process “going in the other direction” if u_1, v_1 are adjacent. □

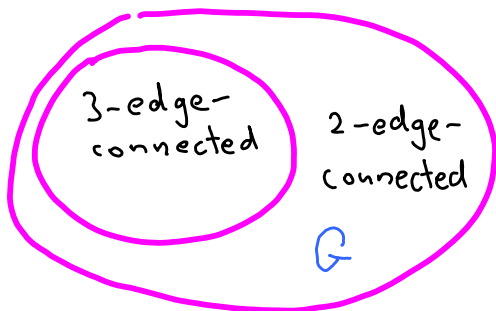
(The next theorem holds by the 4CT, but we are proving it without using 4CT.)

Theorem 4.4.3. Every 2-edge-connected 3-regular simple planar graph is 3-edge-colourable if and only if every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable.

Proof Since every 3-edge-connected graph is also 2-edge-connected, we only need to show that if every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable, then every 2-edge-connected 3-regular simple planar graph is 3-edge-colourable. So suppose that every 3-edge-connected 3-regular simple planar graph is 3-edge-colourable, and let G be a 2-edge-connected 3-regular simple planar graph. We need to show that G is 3-edge-colourable. The proof of this is by induction on $n(G)$.

k -edge-connected $\leftrightarrow$ edge connectivity $\geq k$.

So k -edge-connected $\rightarrow (k-1)$ -edge-connected.



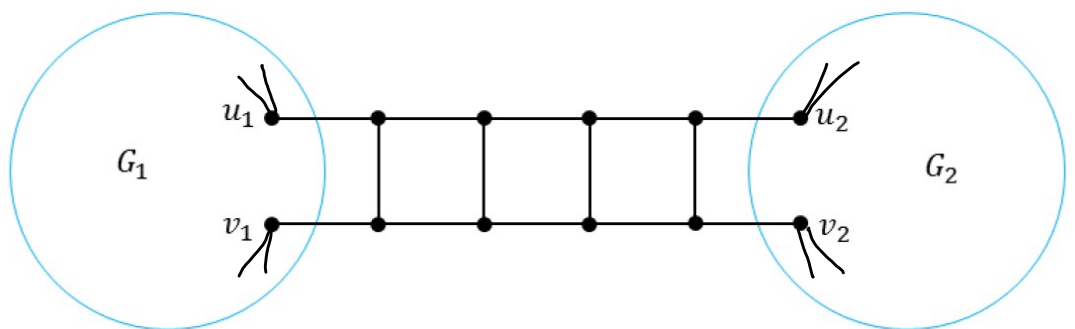
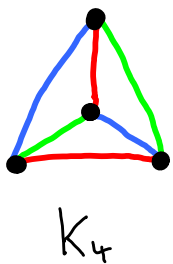
Need:-

Every 3-edge-connected is 3-edge-colourable

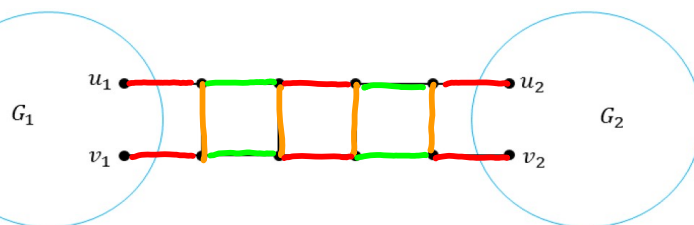
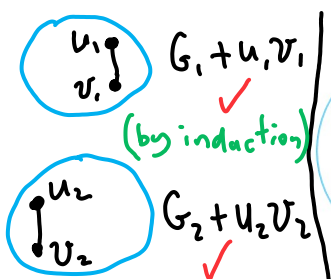


Every 2-edge-connected is 3-edge-colourable

For $n(G) \leq 4$, the only graph to consider is K_4 , which is 3-edge-colourable. Thus, for the inductive step we have $n(G) > 4$. If G is 3-edge-connected, then it is 3-edge-colourable by assumption. So we can assume that G has edge connectivity 2. Thus, by Lemma 4.4.2, G consists of vertex-disjoint subgraphs G_1 and G_2 , non-adjacent vertices $u_1, v_1 \in V(G_1)$, non-adjacent vertices $u_2, v_2 \in V(G_2)$, and a ladder joining (u_1, v_1) to (u_2, v_2) . Let the ladder consist of the paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ and the edges $u'_1 v'_1, u'_2 v'_2, \dots, u'_t v'_t$.



Both $G_1 + u_1 v_1$ and $G_2 + u_2 v_2$ are 2-edge-connected and 3-regular. So by the inductive hypothesis they are 3-edge-colourable. Colour the rungs (edges $u'_1 v'_1, u'_2 v'_2, \dots, u'_t v'_t$) of the ladder with colour c_3 and alternately colour the edges of the paths $[u_1, u'_1, u'_2, \dots, u'_t, u'_{t+1} = u_2]$ and $[v_1, v'_1, v'_2, \dots, v'_t, v'_{t+1} = v_2]$ of the ladder with c_1 and c_2 , beginning by colouring the edges $u_1 u'_1$ and $v_1 v'_1$ with colour c_1 . Thus, the ladder is properly 3-edge-coloured.

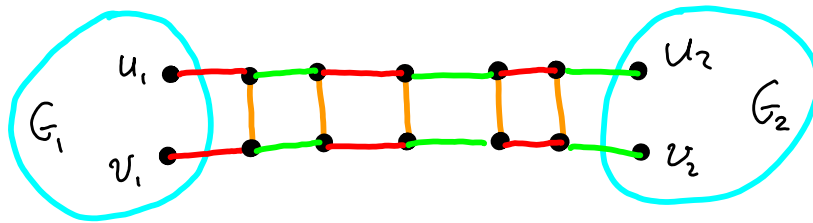


$c_1 = \text{red}$

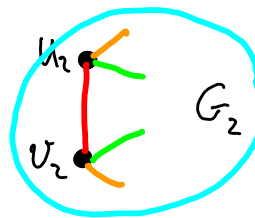
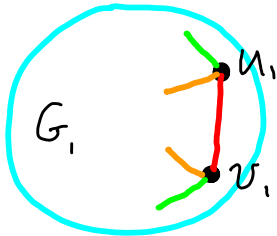
$c_2 = \text{green}$

$c_3 = \text{orange}$

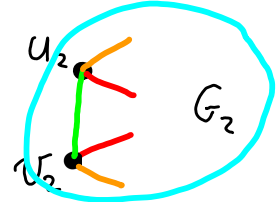
OR



By permuting the colours in a proper 3-edge colouring of $G_1 + u_1v_1$ with colours c_1, c_2, c_3 , we can ensure that the edge u_1v_1 is assigned colour c_1 . Thus, the union of G_1 and the ladder is properly 3-edge-coloured. Similarly, by permuting the colours in a proper 3-edge colouring of $G_2 + u_2v_2$ with colours c_1, c_2, c_3 , we can ensure that the edge u_2v_2 is assigned the same colour as the edges $u'_t u_2$ and $v'_t v_2$ (namely colour c_1 if t is even, and colour c_2 if t is odd). This produces the required proper 3-edge colouring of G . $\square$



OR

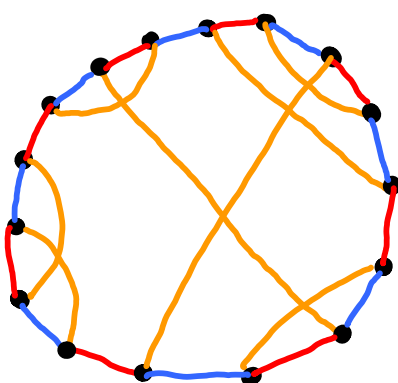


Theorems 4.3.2 and 4.4.3 tell us that the Four Colour Theorem is equivalent to the statement that every simple 3-edge-connected 3-regular planar graph is 3-edge-colourable. We also know that a 3-regular graph (with more than two vertices) is 3-edge-connected if and only if it is 3-connected (Theorem 3.0.5). So the Four Colour Theorem is also equivalent to the statement that every simple 3-connected 3-regular planar graph is 3-edge-colourable.

$\kappa(G) \leq \kappa'(G)$ in general
but $\kappa(G) = \kappa'(G)$ for 3-regular

In 1880, Tait conjectured that every simple 3-connected 3-regular planar graph is Hamiltonian. It is easily seen that a Hamiltonian 3-regular graph is 3-edge-colourable. So Tait's conjecture, if true, implies the Four Colour Theorem.

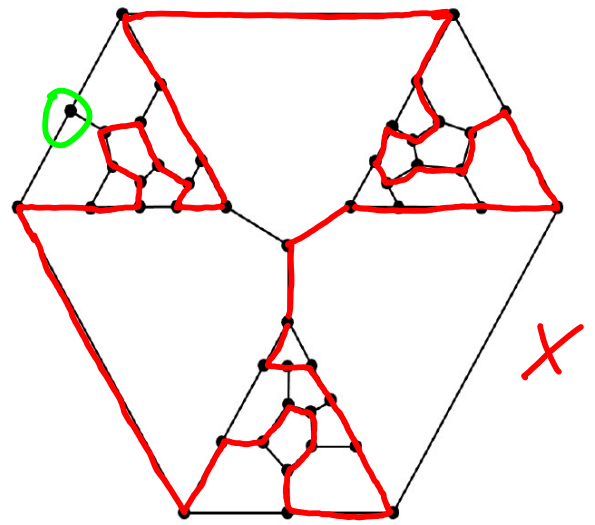
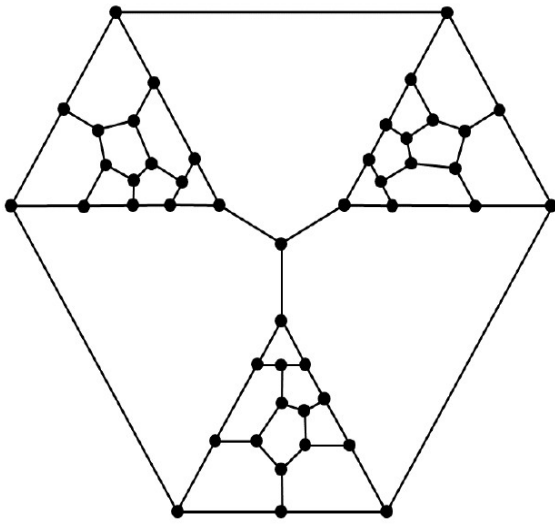
Exercise: Show that a Hamiltonian 3-regular graph is 3-edge-colourable.



3-regular implies
even order.

It wasn't until 1946, that a counter-example to Tait's conjecture was found. This counter-example was found by Tutte [17] and is now known as the *Tutte graph*.

Definition 4.4.4. The **Tutte graph** is the graph shown below.



The following theorem of Grinberg [7] can be used to prove that the Tutte graph has no Hamilton cycle.

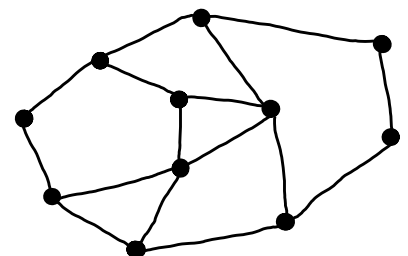
Theorem 4.4.5. (Grinberg's Theorem) Let G be a loopless plane graph with a Hamilton cycle C , let r_i denote the number of faces of length i in the interior of C , and let r'_i denote the number of faces of length i in the exterior of C . Then

$$\sum_i r_i(i-2) = \sum_i r'_i(i-2) = n(G) - 2.$$

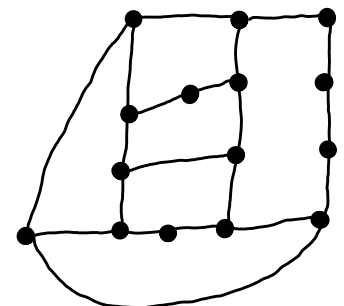
$$\sum_{f \in F_{\text{int}}} (\text{len}(f) - 2) \quad \sum_{f \in F_{\text{ext}}} (\text{len}(f) - 2)$$

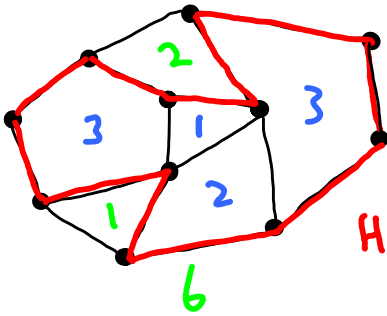
where $F_{\text{int}}/F_{\text{ext}}$ is the set of faces in the interior/exterior of C .

Exercise: Find a Hamilton cycle C in the graph at right and verify that Theorem 4.4.5 holds for C .



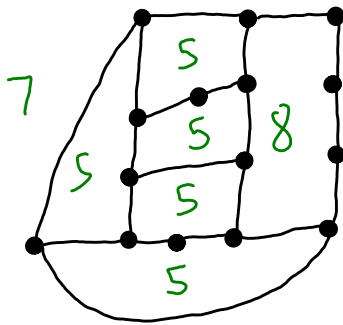
Exercise: Use Grinberg's Theorem to show that the graph at right has no Hamilton cycle.



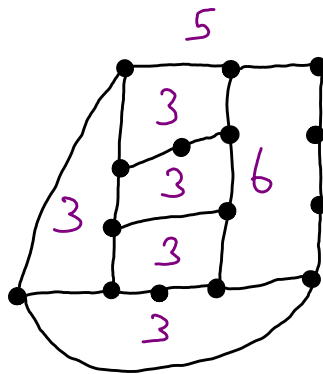


$$\sum_i r_i(i-2) = \sum_i r'_i(i-2) = n(G) - 2.$$

$$1+2+3+3 = 1+2+6 = 11-2 \quad \checkmark$$



face lengths



face lengths - 2

$$\sum_{f \in F_{\text{ext}}} \equiv 2 \pmod{3}$$

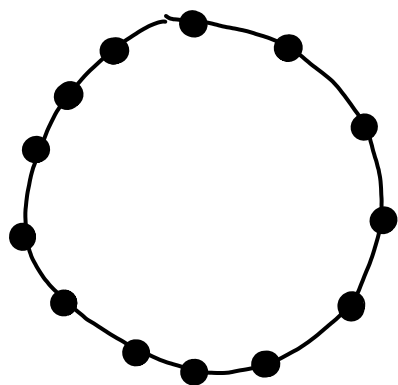
$$\sum_{f \in F_{\text{int}}} \equiv 0 \pmod{3}.$$

Theorem 4.4.5. (Grinberg's Theorem) Let G be a loopless plane graph with a Hamilton cycle C , let r_i denote the number of faces of length i in the interior of C , and let r'_i denote the number of faces of length i in the exterior of C . Then

$$\sum_i r_i(i-2) = \sum_i r'_i(i-2) = n(G) - 2.$$

Proof We begin by showing that $\sum_i (i-2)r_i = n(G) - 2$. The proof of this is by induction on the number t of edges in the interior of C . If there are no edges in the interior of C , then the result holds. Thus, we have $t \geq 1$ for the inductive step.

Let $n = n(G)$.



No edges inside C

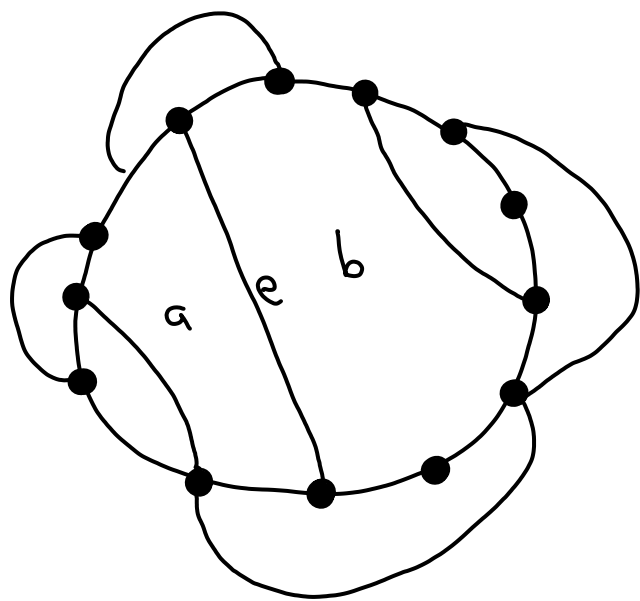
$\Rightarrow r_n = 1, r_i = 0$ for $i \neq n$

So $\sum_i r_i(i-2) = n-2$. ✓

Let G' be the graph obtained from G by deleting an edge e from the interior of C , and suppose that in G the two faces on either side of e have lengths a and b . In G' , instead of these two faces we have a single face of length $a + b - 2$ (and all the other faces are unchanged). Thus, by the inductive hypothesis, and since $\underbrace{(a-2)}_{\text{with } e} + \underbrace{(b-2)}_{\text{without } e} = \underbrace{(a+b-2)-2}$, we have $\sum_i (i-2)r_i = n(G) - 2$.

with e

without e



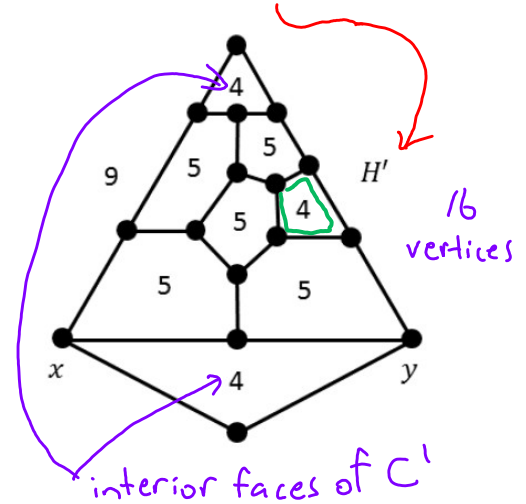
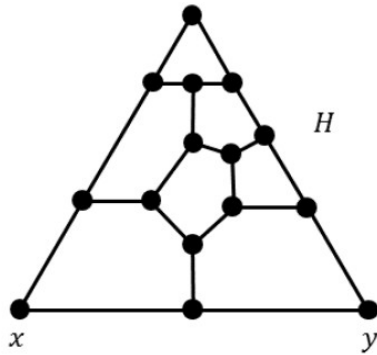
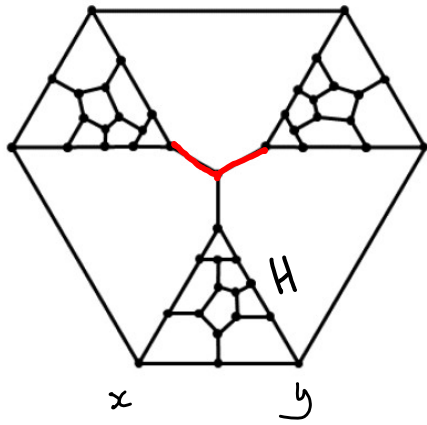
$\left. \begin{matrix} a=6 \\ b=7 \end{matrix} \right\}$ deleting $e \Rightarrow$ single face of length 11

e is on the boundary of two faces, so the length of the single face is 2 less than the sum of the lengths of the original two faces.

There is no fundamental difference between the interior and the exterior of C (consider an embedding on a sphere rather than in the plane). So we also have $\sum_i (i-2)r'_i = n(G) - 2$. $\square$

Theorem 4.4.6. The Tutte graph is non-Hamiltonian.

Proof For a contradiction, suppose C is a Hamilton cycle in the Tutte graph. Consider the three isomorphic components that remain if the central vertex and the three longest edges of the Tutte graph (as shown below) are deleted. Let H be the component that does not contain an endpoint of the two edges of C that are incident with the central vertex. The edges of C that are in H form a Hamilton x, y -path in H where x and y are the two vertices as indicated below. Thus, the graph H' obtained from H by adding a new vertex that is joined to x and y has a Hamilton cycle C' .



Let z be the number of 5-faces in the exterior of C' .

$$3z + 7 = 14 \quad \text{or} \quad 3z + 7 + 2 = 14 \quad \left. \vphantom{3z + 7 = 14} \right\} \text{contradiction}$$

$$3z = 7 \quad \text{or} \quad 3z = 5$$

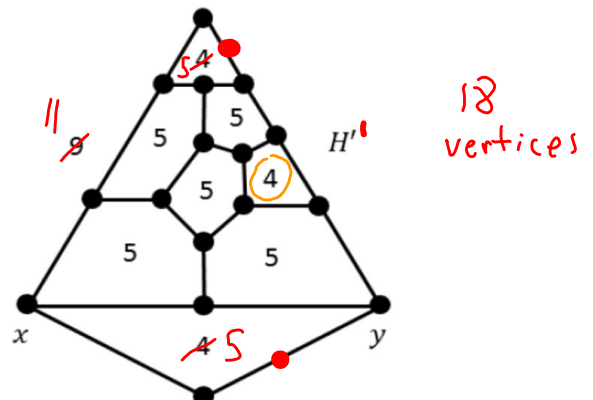
(if the green face is in the interior of C') (if the green face is in the exterior of C')

Alternatively, let H'' be the graph

(H'' is Hamiltonian iff)
(H' is Hamiltonian)

Grinberg $\Rightarrow$ all faces except one contribute $0 \pmod{3}$

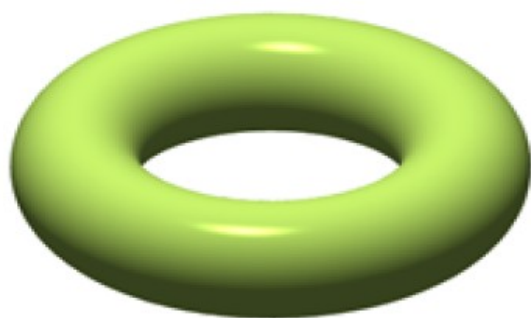
$$n(G) - 2 = 16$$



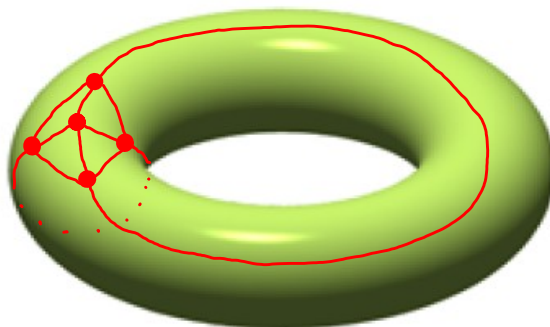
4.5 Embeddings on other Surfaces

An embedding of a graph on an arbitrary surface is defined in a similar way as an embedding of a graph in the plane; the graph is drawn on the surface such that there are no edge-crossings.

A graph can be embedded in the plane if and only if it can be embedded on a sphere. To see this imagine puncturing a face of a graph that embedded on the sphere and then stretching the surface out flat. The punctured face becomes the external face. This process is reversible. We shall see however that there are non-planar graphs that can be embedded on other surfaces.

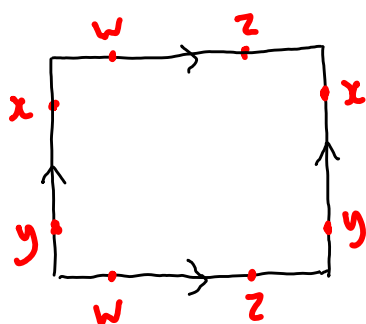


Torus

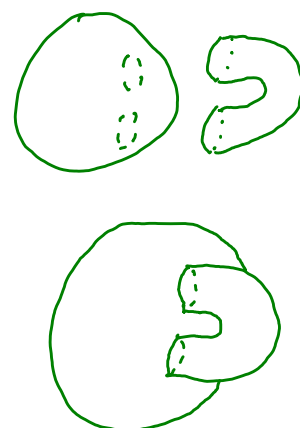


K_5 embedded on the torus.

A **torus**, the familiar “doughnut-shaped” surface, can be constructed in several different ways. One way is to begin with the unit square, identify the point $(x, 0)$ with the point $(x, 1)$ for each x in the interval $[0, 1]$, and identify the point $(0, y)$ with the point $(1, y)$ for each y in the interval $[0, 1]$. Another way to construct a torus is to glue a handle to a sphere. Generalising this construction, the surface obtained from a sphere by the addition of k handles is the **orientable surface of genus k** , and is denoted by S_k .



Torus



Torus: S_1



S_2



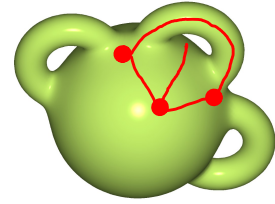
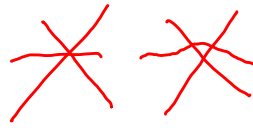
S_3



S_3

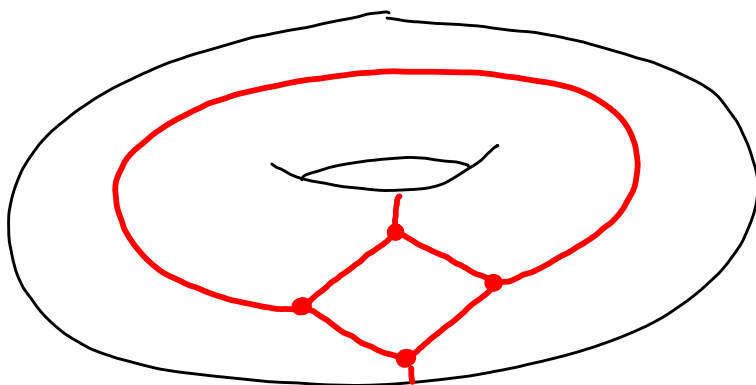
Any graph can be drawn on a sphere such that there is no point where more than two edges cross. At any point where two edges do cross, a handle can be placed so that one of the edges goes around the handle and the crossing is avoided. Thus, any graph can be embedded on some orientable surface. For a given graph G , the smallest non-negative integer k such that G can be embedded on S_k is called the **genus** of G .

What is the genus of K_5 ?

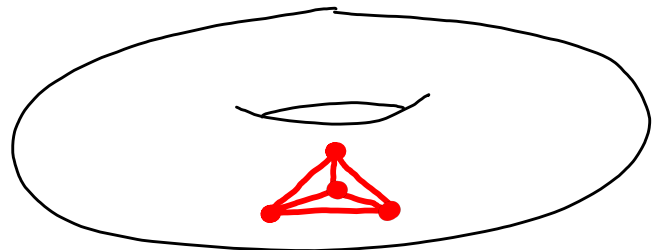


A face f of a graph that is embedded on a surface is a **2-cell** if every closed curve in f can be continuously deformed within that region to a point (that is, a face is a 2-cell if it is homeomorphic to a disk). An embedding of a graph on a surface is a **2-cell embedding** if every face of the embedding is a 2-cell. It can be seen that any embedding of a connected graph on a sphere is a 2-cell embedding. This is not true for other surfaces.

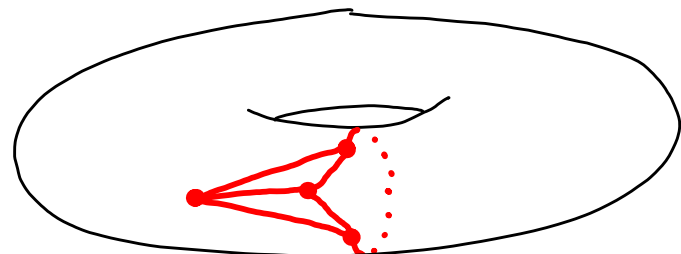
Exercise: Draw two embeddings of K_4 on the torus; one that is a 2-cell embedding and one that is not.



A 2-cell embedding of K_4 on the torus.



Two embeddings of K_4 on the torus that are not 2-cell embeddings



The following theorem generalises Euler's Identity.

Theorem 4.5.1. If a connected graph having n vertices and m edges has a 2-cell embedding on S_k , then the number r of faces satisfies

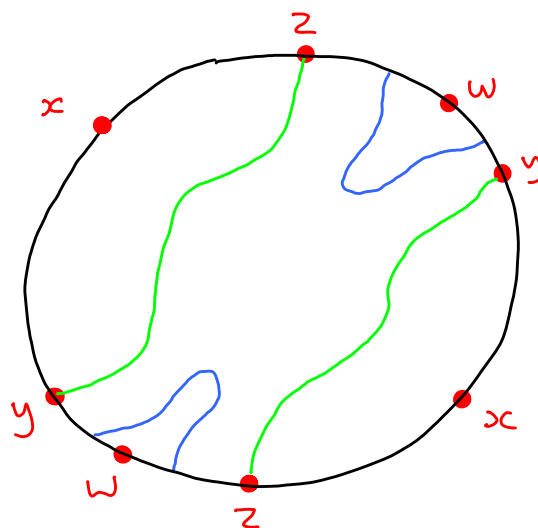
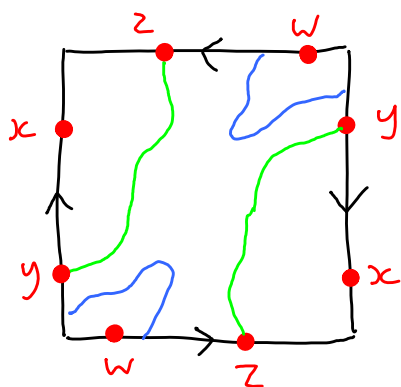
$$n - m + r = 2 - 2k.$$

On the plane/sphere S_0 we have $k=0$ and $n - m + r = 2$.
(Euler's Identity).

A 2-cell embedding of K_4 on the torus has 2 faces.

Theorem 4.5.2. If a graph of genus k is embedded on S_k , then the embedding is a 2-cell embedding.

The *projective plane* is another surface on which one can embed graphs. The projective plane can be constructed from the unit square by identifying the point $(x, 0)$ with the point $(1 - x, 1)$ for each x in the interval $[0, 1]$, and identifying the point $(0, y)$ with the point $(1, 1 - y)$ for each y in the interval $[0, 1]$. Alternatively, the projective plane is obtained from a circular disk by identifying each pair of antipodal points on the circumference.



Theorem 4.5.3. If a connected graph having n vertices and m edges has a 2-cell embedding on the projective plane, then the number r of faces satisfies

$$n - m + r = 1.$$

Shown below is an embedding of the Heawood graph on the torus and an embedding of the Petersen graph on the projective plane. The dual graph of this embedding of the Heawood graph is K_7 and the dual graph of this embedding of the Petersen graph is K_6 .

